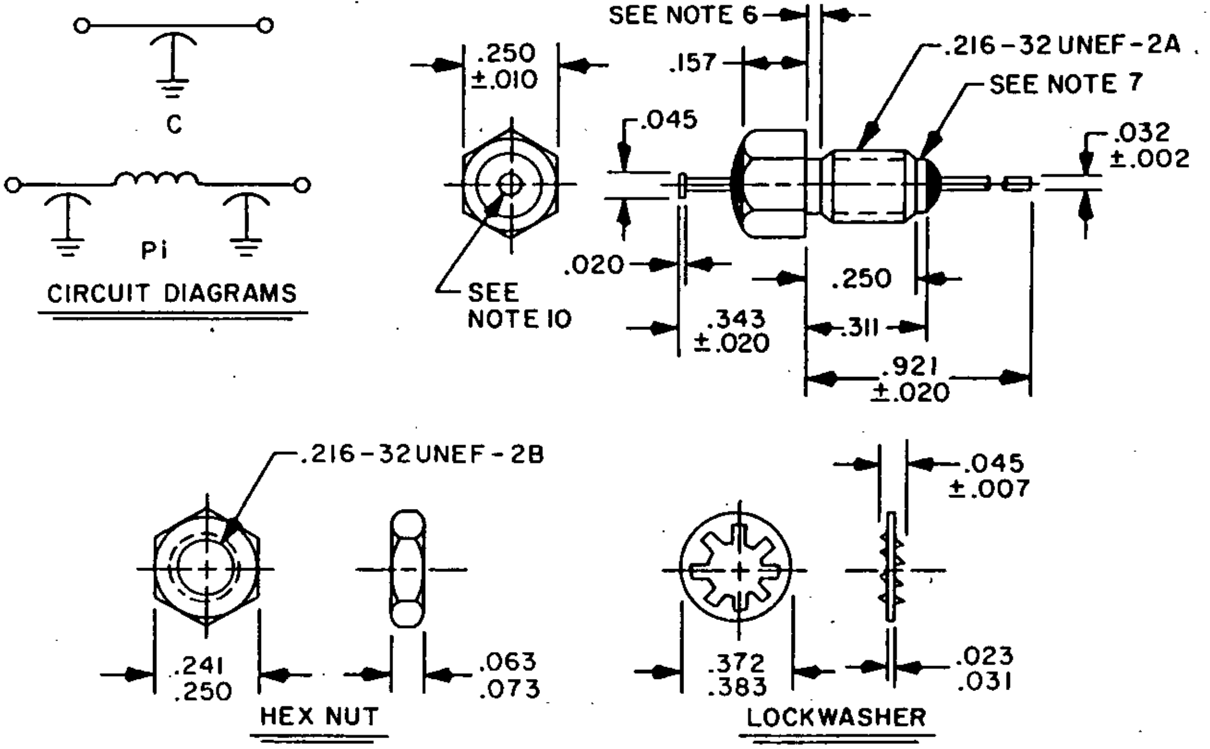
MIL-F-28861/9B
31 December 1987
SUPERSEDING
MIL-F-28861/9A
29 August 1986

MILITARY SPECIFICATION SHEET

FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, NONHERMETICALLY SEALED, STYLE FS63

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-28861.



Inches	mm	Inches	mm	Inches	mm
.002	0.05	.032	0.81	.241	6.12
.005	0.13	.040	1.02	.250	6.35
.007	0.18	.045	1.14	.311	7.90
.010	0.25	.063	1.60	.343	8.71
.020	0.51	.073	1.85	.372	9.45
.023	0.58	.157	3.99	.383	9.73
.031	0.79	.216	5.49	.921	23.39

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005.
4. Circuit diagram is for information only.
5. All filters shall be supplied with mounting hardware (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
6. Imperfect thread or undercut optional .040 inch maximum.
7. One imperfect thread allowed .040 inch maximum.
8. Recommended mounting torque: 64 oz-in. ±4 oz-in.
9. Potting shall not extend .030 inch from the filter body.
10. Turret head terminal optional.

FIGURE 1. Case and hardware dimensions and circuit diagrams.

Ⓑ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 2.5 grams maximum.

Ⓑ Case finish: S only (silver-plated).
In accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 10 amperes, maximum.

Capacitance to ground: See table I for capacitance value.

Dissipation factor: 3.0 percent maximum.

Voltage and temperature limits of capacitance: +15, -40 percent.

Insulation resistance:

At +25°C: 1000 megohm-microfarad or 100,000 megohms minimum, whichever is less.

At +125°C: 100 megohm-microfarad or 10,000 megohms minimum, whichever is less.

Insertion loss: At +25°C: In accordance with table I.
At -55°C and +125°C: A 3 dB degradation from the +25°C value shall be allowed.

Voltage drop: 0.1 volt, maximum.

DC resistance: 0.01 ohm, maximum.

Seal: Not applicable.

Temperature rise: +25°C, maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Ⓑ Product assurance level: Class B only.

Ⓑ Part number: The part number shall be as follows:

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M28861/09-     XXX            S                B
                |             |                |
           Dash no.      Case finish   Product assurance
           per figure 1                level per MIL-F-28861
           and table I
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Ⓑ TABLE I. Electrical characteristics.

Dash number	Circuit	Rated voltage		Capacitance (µF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/					
		DC volts	AC 3/ volts		1 MHz	10 MHz	100 MHz	200 MHz	1 GHz	10 GHz
001	Pi	50	-	.12	20	38	58	70	70	70
002	C	50	-	.20	26	36	52	60	70	70
003	C	100	-	.05	15	35	38	50	70	70
004	C	100	-	.10	20	38	48	53	70	70
005	Pi	100	-	.075	8	38	70	70	70	70
006	C	200	115	1,000 pF	-	4	20	25	40	50
007	C	200	115	5,000 pF	-	15	34	41	45	50
008	C	200	115	.01	4	21	35	42	55	60
009	Pi	200	-	.012	-	16	65	65	70	70
010	C	200	115	.025	8	28	36	44	64	70

Ⓑ 1/ For C circuits, insertion loss measurements shall be made under no load. For Pi circuits, insertion loss measurements shall be made under full load over the frequency range of 1 MHz to 10 MHz; above this frequency range, insertion loss shall be measured under no load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

3/ 0 to 400 Hz.

Marking: Filters shall be marked, as a minimum, with an abbreviated military part number as shown on figure 2. Full marking, in accordance with MIL-F-28861, shall be marked on the unit package.

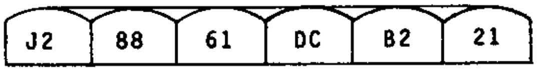

FIGURE 2. Example of marking for the abbreviated military part number on the hex flats - expanded view.

J - Abbreviation for JAN
28861 - Abbreviation for M28861
D - Indicates specification sheet 09
C - Code indicating dash no. as follows:

Code	Dash no.	Code	Dash no.
A	001	F	006
B	002	G	007
C	003	H	008
D	004	J	009
E	005	K	010

B - Code indicating product assurance level (for example, B indicates class B).
221 - Date code. First digit indicates the year (for example, 2 indicates 1982). The last two digits indicate the week.

Cataloging information: Dash numbers 002 through 004, 006 through 008, and 010 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 001, 005, and 009 shall be cataloged under FSC 5915 as radio frequency interference filters.

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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